

MVR Rev. A2 vs. Cesium: Five-Day Phase Run Analysis

Summary

- **Frequency offset:** $+2.70 \times 10^{-13}$ versus the Cs. Correcting for the Cs's separately measured offset of -2.8×10^{-13} versus GPS, the MVR is about **-1×10^{-14} versus GPS**, which is zero within a few $\times 10^{-14}$.
- **Diurnal effect:** none detectable in five days. A 24-hour component of about 4 ns is fitted, but it is no stronger than fits at neighboring periods (22-30 h), so it is indistinguishable from background wander.
- **Dominant feature:** irregular phase wander of ± 20 -25 ns over periods of hours (12 ns RMS, 86 ns peak-to-peak after removing the linear trend).
- **Drift:** no meaningful long-term frequency drift (about -2×10^{-14} per day from a quadratic fit, which is noise-level).

Data

Item	Value
File	mvr_a2_5_day.csv
Samples	4,320,000 at 10 Hz (0.1 s spacing), no gaps
Duration	432,000 s (exactly 5.000 days)
Columns	time (s), phase (s)
Sign convention (assumed)	phase = MVR minus Cs, so rising phase means the MVR is running fast

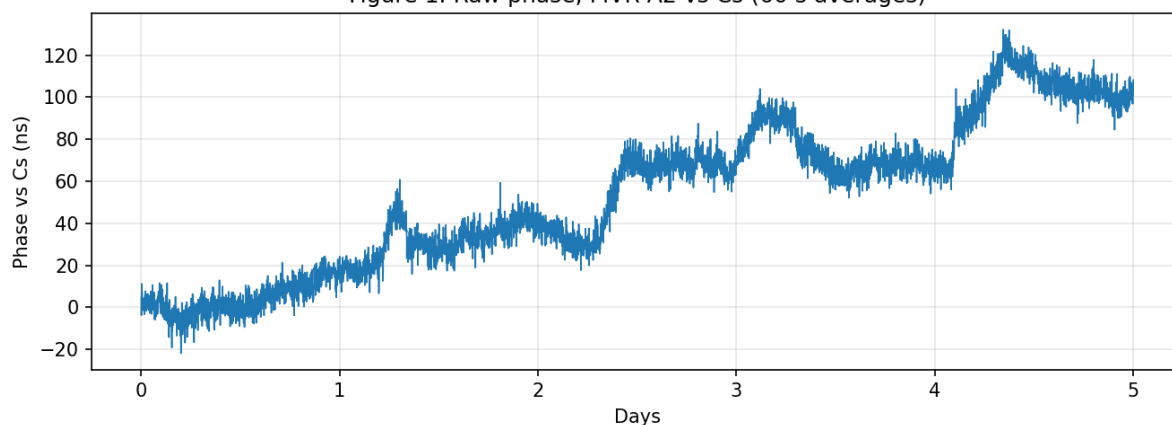
The sign assumption is supported by a sanity check: the opposite convention would put the MVR at -5.5×10^{-13} versus GPS, which is implausible for a disciplined oscillator.

Frequency offset

Measure	Value
Linear fit slope, full run	$+2.70 \times 10^{-13}$
Endpoint-to-endpoint slope	$+2.63 \times 10^{-13}$
Cs offset versus GPS (separately measured)	-2.8×10^{-13}
MVR versus GPS (derived)	about -1×10^{-14}
Quadratic (drift) term	about -2×10^{-14} per day

Per-day slopes versus the Cs vary from -2.7×10^{-13} to $+5.5 \times 10^{-13}$ (days 1-5: +2.2, +2.4, +5.5, -2.7, $+2.5 \times 10^{-13}$). That spread reflects the multi-hour phase wander described below, not real frequency steps. The phase wander alone limits the full-run slope to roughly $\pm 3 \times 10^{-14}$, so the derived offset versus GPS is consistent with zero.

Figure 1. Raw phase, MVR A2 vs Cs (60 s averages)



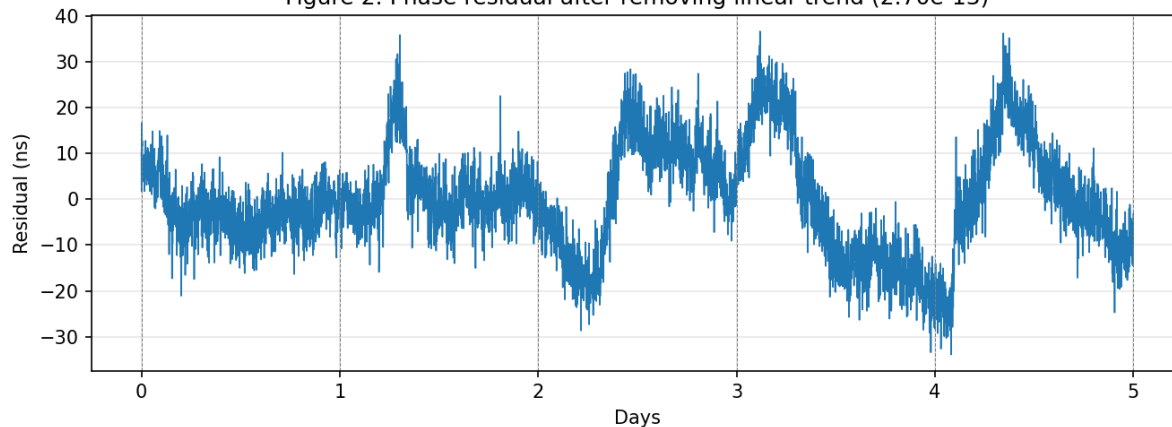
Phase wander

After removing the linear trend the residual phase is **12 ns RMS and 86 ns peak-to-peak**. It is broadband, slow wander rather than a clean cycle:

- Day 1 has a sharp bump of about +13 ns around hours 6–8.
- Day 2 swings from about −18 ns to +18 ns.
- Day 3 runs from about +24 ns to −21 ns.
- Day 4 runs from about −23 ns to +25 ns.

Hourly-average frequency scatters about $\pm 25 \times 10^{-13}$ (1σ). This is measurement and wander noise, consistent with the ADEV at 1 hour, and not evidence of frequency steps.

Figure 2. Phase residual after removing linear trend (2.70e-13)

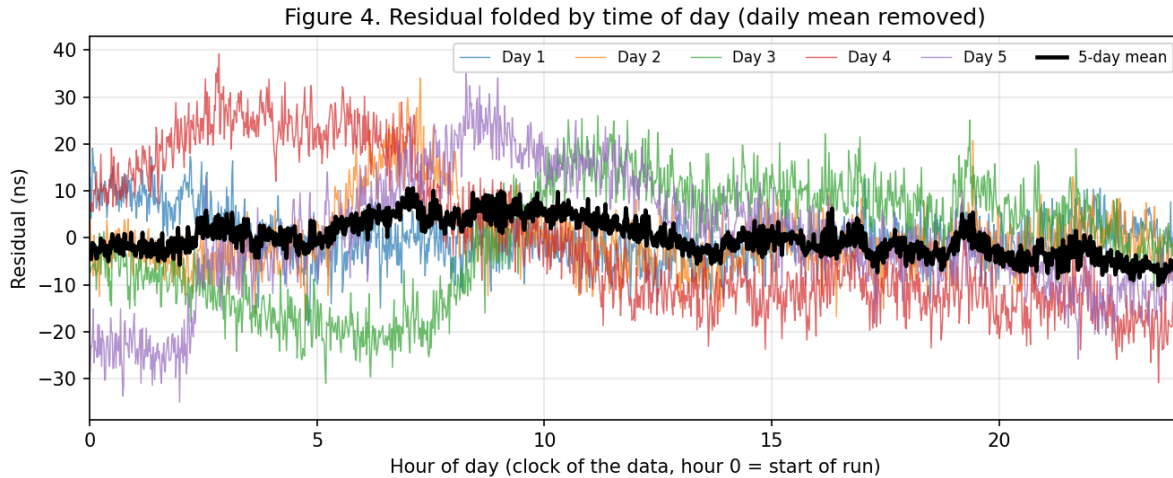


Diurnal check

- A 24-hour sine fit to the phase gives about 4.3 ns, and a 12-hour component of about 1.3 ns.
- The 24-hour fit is **not significant**: fits at 22, 23, 25, 26 and 30 hours give 4.0–4.5 ns as well, so 24 h does not stand out from the background.
- The periodogram of the residual has its largest peaks at 40 h, 15 h and 60 h, with nothing special at 24 h.
- Folding the residual by time of day (Figure 4) shows a hint of a +5 ns hump around hours 6–10, but the day-to-day scatter at those hours is 10–15 ns, larger than the hump.

- **Verdict:** nothing diurnal is detectable. With only five cycles, a diurnal term below about 4–5 ns (under 1×10^{-13} in frequency) cannot be ruled out; ten days or more would help.

Note: the hour of day in Figure 4 is counted from the start of the run. The wall-clock start time of the data is not recorded in the file.

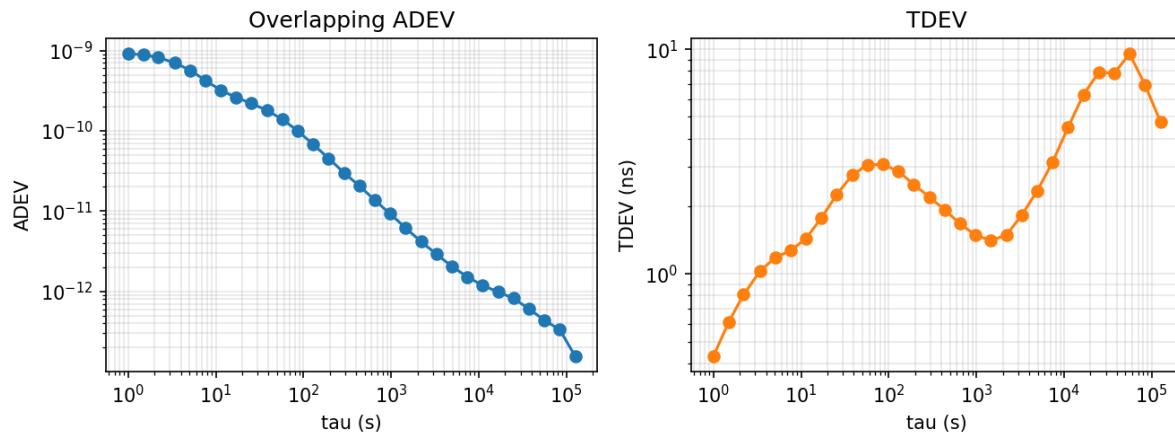


Stability

τ (s)	ADEV	TDEV (ns)
1	9.1×10^{-10}	0.43
10	3.5×10^{-10}	1.4
100	8.7×10^{-11}	3.0
1,000	9.0×10^{-12}	1.5
10,000	1.2×10^{-12}	4.1
100,000	2.1×10^{-13}	5.7

- The 1 s figure is close to the noise floor of the measurement. ADEV falls roughly as $1/\tau$ out to about 1,000 s, which is the signature of white phase noise.
- TDEV has a bump of about 3 ns at 50–100 s, then dips to about 1.5 ns at 1,000–2,000 s. This is typical of a loop correcting GPS timepulse noise at short τ and then following GPS and the environment at longer τ .
- TDEV rises again beyond about 5,000 s, reflecting the multi-hour wander above.
- The data alone cannot say whether that wander originates in the GPS solution, the VCXO's environment, or the Cs and measurement side. A second GPSDO or logged temperature over the same period would help separate these.

Figure 3. Stability



Method notes

- Phase data were analyzed at the full 10 Hz rate for ADEV/TDEV (overlapping estimators, allantools) and block-averaged to 60 s for the trend, fold, sine-fit and periodogram analyses.
- Hourly and ten-minute frequencies were computed from phase differences.
- Sine fits were least-squares, with offset and slope fitted simultaneously.